Rajagopal Somaskandan

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Summary

M.Sc. Computer Science (AI/ML) student with hands-on experience in Machine Learning, Generative AI, NLP, Computer Vision, and RAG systems through internships, IEEE publications, and hackathon projects.

Education

Sathyabama Institute of Science and Technology Aug 2025 – Jun 2027 (Expected)
M.Sc. Computer Science *Chennai, India*

Sathyabama Institute of Science and Technology 2025
B.Sc. Computer Science with AI – ***First Class*** *Chennai, India*

Technical Skills

Programming & Data: Python, SQL, NumPy, Pandas, Git, GitHub
Machine Learning: Scikit-learn, XGBoost, LightGBM, Feature Engineering, Learning-to-Rank, Model Evaluation, Explainable AI (SHAP)
Deep Learning & Computer Vision: PyTorch, CNNs, OpenCV, 3D Gaussian Splatting, COLMAP, FFmpeg
GenAI & Agentic AI: Hugging Face Transformers, RAG, LangChain, LangGraph, MCP, LoRA/PEFT, Fine-tuning, Sentence-Transformers (SBERT), FAISS, ChromaDB
Backend & APIs: FastAPI, React, REST APIs, PostgreSQL, MongoDB
Production AI & MLOps: Docker, MLflow, CI/CD, Kubernetes, AWS, Linux

Projects

SourceUp: Explainable Supplier Ranking System Research Project – Submitted for IEEE
- Built a **supplier recommendation system** using **SBERT**, **FAISS**, and **LightGBM LambdaRank** over **1.27M supplier records**.
- Created a weak-supervision benchmark with **23K+ supplier–query pairs**, achieving **NDCG@10 of 0.886**; evaluated with **MAP** and **MRR**.
- Integrated **SHAP** explainability and fairness-aware ranking for transparent recommendations.

FinGuard Pro: Explainable Fraud Detection Published, IEEE ICAAIC 2025
- Developed a real-time **fraud detection pipeline** using **XGBoost**, **LightGBM**, and **CatBoost** on **100K transactions**, achieving **94% recall** and **ROC-AUC of 0.9951** with **SHAP**-based explanations.

S.A.F.E: Privacy-Preserving Crowd Anomaly Detection Design Thinking Project, 2026
- Evaluated **five anomaly detection models** across **three benchmark datasets** (ETH/UCY, Zara1/Zara2, Mall), selecting **Median Absolute Deviation (MAD)** for best cross-dataset generalization at **ROC-AUC 0.8391** while cutting false positives by **40–60%**.
- Built a privacy-preserving crowd anomaly detection system with **FastAPI** and real-time explainable alerts.

Experience

Production AI Developer (Intern) Mar 2026 – Jun 2026
InnoXR Labs Pvt. Ltd. *Chennai, India*
- Built an end-to-end **ML pipeline** for single-video **3D reconstruction** using **COLMAP**, **Gaussian Splatting**, and **PyTorch** (MonoSplat), with **GPU-accelerated preprocessing** for large-scale video processing.
- Built **FastAPI** backend services and integrated a **Three.js** viewer for real-time 3D visualization; benchmarked on Tanks and Temples.

Data Analyst Intern Jul 2024 – Oct 2024
India Advocacy *Chennai, India*
- Built **Java and Selenium ETL pipelines** processing over **1M records**, including large-scale data cleaning and preprocessing.

Achievements & Hackathons

Best Industry Innovation Award, Industry Innovation '26 – National-level hackathon, Sathyabama Institute of Science and Technology, competing against **63+ teams** (Mar 2026)

4th Place, Origin 24-Hour Hackathon – SIMATS Engineering, Chennai (Apr 2026)

Certifications

AI & ML | *Space Zee Technologies* Jul 2024
Python Core | *Space Zee Technologies* Feb 2024